Introductory Biology (BT 101)

Why ?

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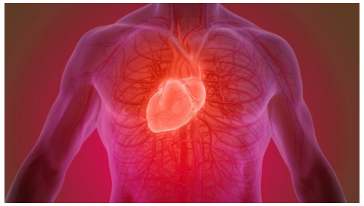
Some Remarkable Facts of Biology

Human = Machine + Think



Heart (marvelous pump)

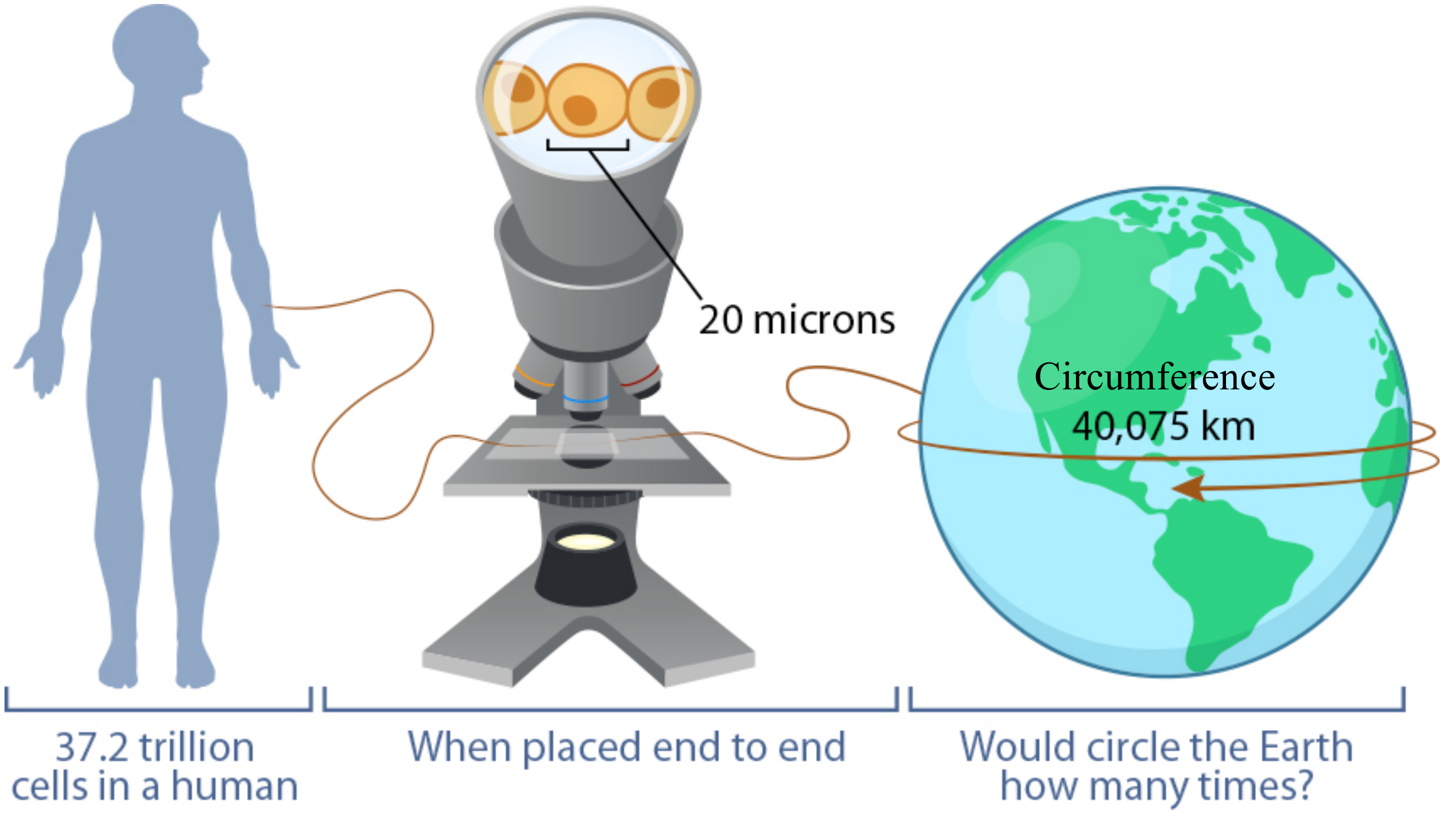
Wt~ 300 gm

- Made up of Proteins+fat+water.

- Size of your heart = your two clenched fists

- 70 times/min = 100,000 times/day (7500 liters/day). Fuel tank of a Car could be filled within 7-8 minutes by heart.
- Blood is circulated through 96,000 Km blood vessel each day. (Go around the world more than TWICE).
- Regulate speed and faster distribution during exercise.
- No noise. Non stop for 70-80 years

Can we build this kind of machine ?



1 Trillion = 10^{12}

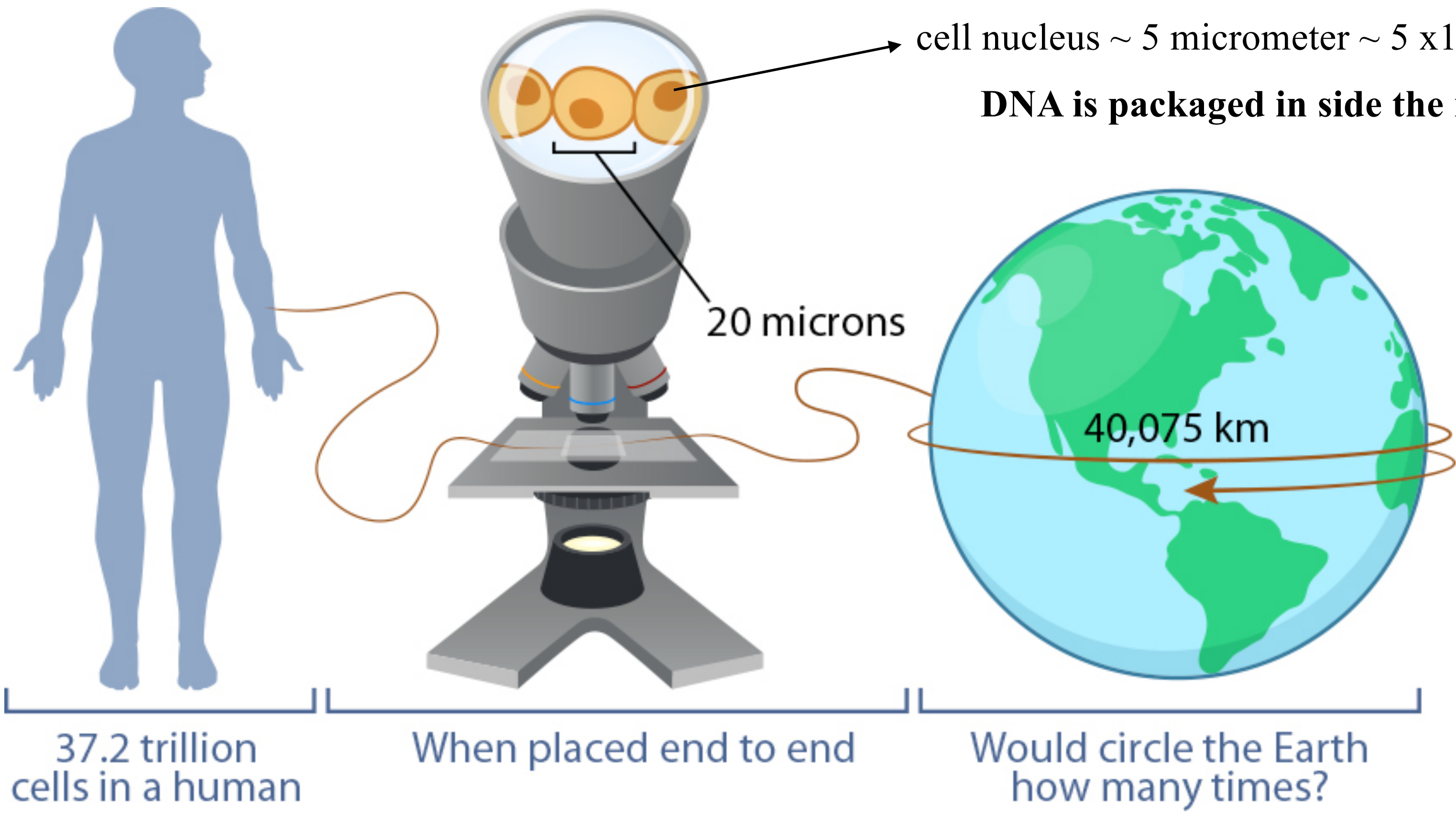
1 micron = 10^{-9} km

$[\,(37.2 \text{x} 10^{12}) \text{x}\ (20 \text{x}\ 10^{-9})\,]\ /\ 40075$

= **18.6 Times**

cell nucleus ~ 5 micrometer ~ 5 x10^{-6} meter

DNA is packaged in side the nucleus



How the DNA is packaged ?

http://www.sciencephoto.com/media/209697/view

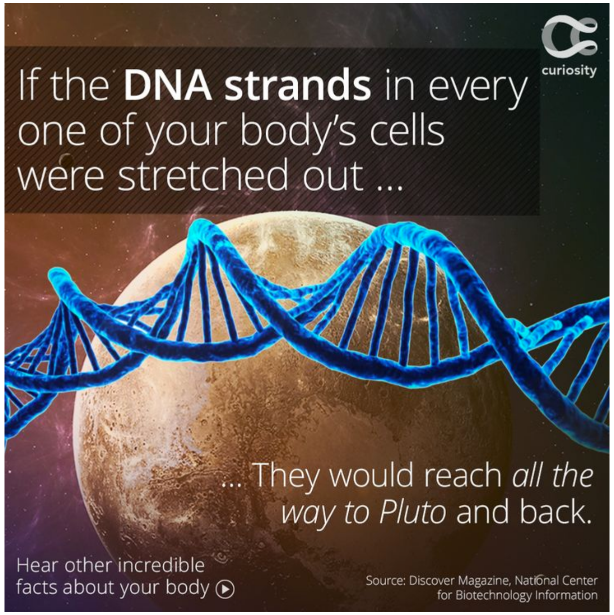
Coloured TEM of DNA from E. coli bacterium

The total length of DNA is 1.5 millimetres, 1000 times the length of the bacterium itself.

How long is our DNA if we stretch it out?

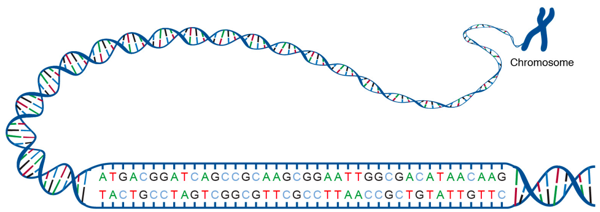
Ans: DNA in one cell ~ **2m** long

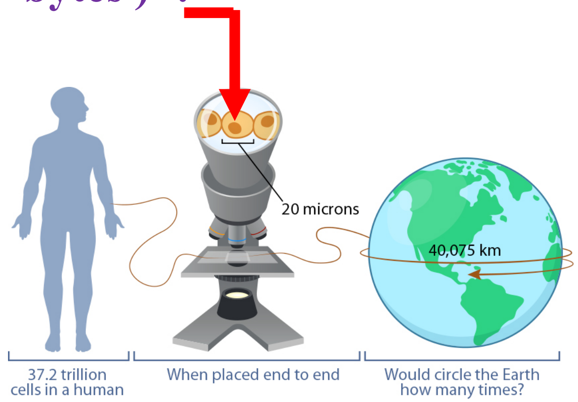
Packed in side nucleus ~ **$5x10^{-6}$** m



Can we build a packaging company similar to that ?

How much information in a single human cell (genome, Base pairs => bytes) ?





- bit = 0 or 1
- 8 bits = 1 byte
- Single base in binary codes (AT = 00, TA=10, GC=01, CG = 11)
- 1 base pair = 2 bits = ¼ byte
- Size of single human genome = **$6x10^9$** base pairs = $(6x10^9 /4)$ bytes ~$1.5x10^9$ byte =1.5GB

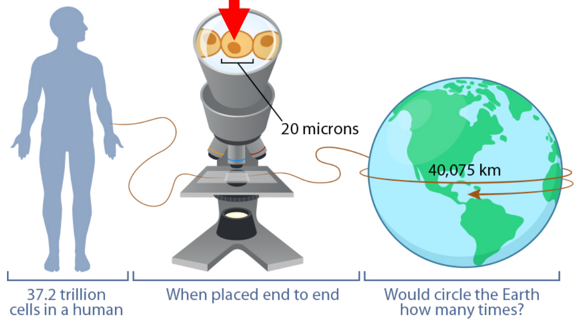
1 human genome = 1.5 GB data

World population = 7.8 billion as of January 2021
= $(7.8x10^9)$x 1.5 GB
= 11.7×10^9 GB

Instructions to make proteins

6×10^9 base pairs

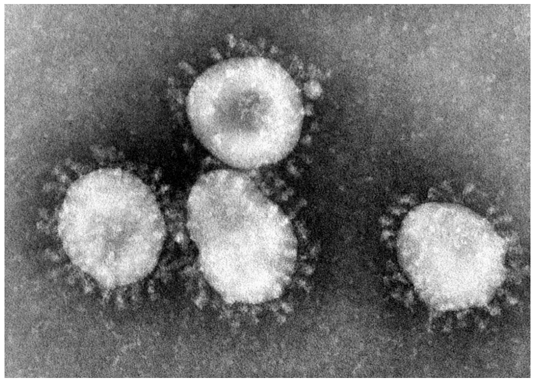
Same is true for microbes (e.g, bacteria)



Can we explore the difference (Say bacterial vs human proteins) ?

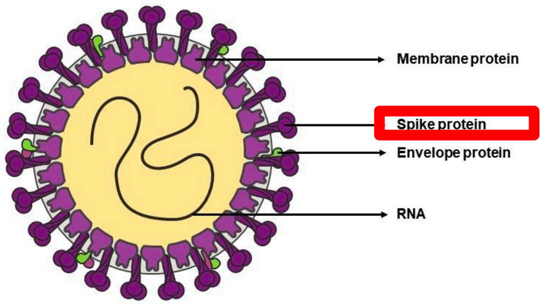
DRUGS

- RNA
- Membrane enclosed
- Corona "CROWN" is spike protein



https://en.wikipedia.org/wiki/Coronavirus

COVID 19 Virus Labeled Diagram with Membrane Protein



DRUG-Target

https://www.slideteam.net/covid-19-virus-labeled-diagram-with-membrane-protein.html

Nature **594, 88–93 (2021).** https://doi.org/10.1038/s41586-021-03491-6

Article

Drugs that inhibit TMEM16 proteins block SARS-CoV-2 spike-induced syncytia

https://doi.org/10.1038/s41586-021-03491-6

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Check for updates

Luca Braga[1,9], Hashim Ali[1,9], Ilaria Secco[1], Elena Chiavacci[1], Guilherme Neves[2,3], Daniel Goldhill[4], Rebecca Penn[4], Jose M. Jimenez-Guardeño[5], Ana M. Ortega-Prieto[5], Rossana Bussani[6], Antonio Cannatà[1], Giorgia Rizzari[1], Chiara Collesi[6,7], Edoardo Schneider[1,7], Daniele Arosio[8], Ajay M. Shah[1], Wendy S. Barclay[4], Michael H. Malim[5], Juan Burrone[2,3] & Mauro Giacca[1,6,7]✉

COVID-19 is a disease with unique characteristics that include lung thrombosis[1], frequent diarrhoea[2], abnormal activation of the inflammatory response[3] and rapid deterioration of lung function consistent with alveolar oedema[4]. The pathological substrate for these findings remains unknown. Here we show that the lungs of patients with COVID-19 contain infected pneumocytes with abnormal morphology and frequent multinucleation. The generation of these syncytia results from activation of the SARS-CoV-2 spike protein at the cell plasma membrane level. On the basis of these observations, we performed two high-content microscopy-based screenings with more than 3,000 approved drugs to search for inhibitors of spike-driven syncytia. We converged on the identification of 83 drugs that inhibited spike-mediated cell fusion, several of which belonged to defined pharmacological classes. We focused our attention on effective drugs that also protected against virus replication and associated cytopathicity. One of the most effective molecules was the antihelminthic drug niclosamide, which markedly blunted calcium oscillations and membrane conductance in spike-expressing cells by suppressing the activity of TMEM16F (also known as anoctamin 6), a calcium-activated ion channel and scramblase that is responsible for exposure of phosphatidylserine on the cell surface. These findings suggest a potential mechanism for COVID-19 disease pathogenesis and support the repurposing of niclosamide for therapy.

Technology developments

Can we use bacteria to produce useful HUMAN proteins ?

Advantage : bacteria can **grow fast** and produce **large amounts** of these **useful proteins. (**e.g, insulin, Growth hormone **)**

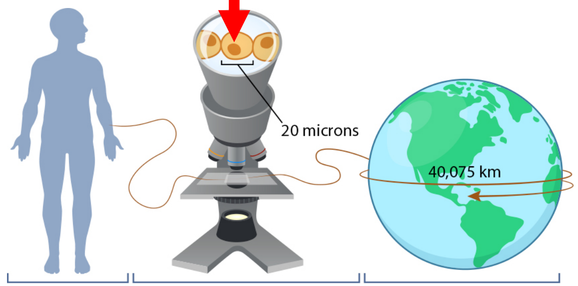
Herbert Boyer

E. coli bacterium

1**976**: **Genentech**

1978: produced synthetic **insulin** using his new transgenic genetically modified **bacteria**
1979: Human growth hormone.

Mutation causes disease ← Instructions to make proteins ← **$3x10^9$ base pairs**



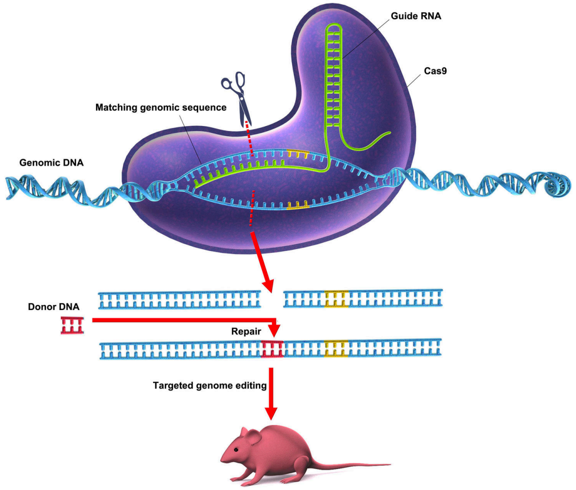
Can we FIX the mutation in the DNA in the embryo ?

Under active investigations….

Emmanuelle Charpentier *Jennifer Doudna*

CRISPR-Cas9 editing technique

Nobel Prize in Chemistry (Year 2020)



https://news.cnrs.fr/articles/genetic-scissors-for-tailored-neuroscience-solutions

https://www.youtube.com/watch?v=9q8AGst7KiY

Introductory Biology
Why ?

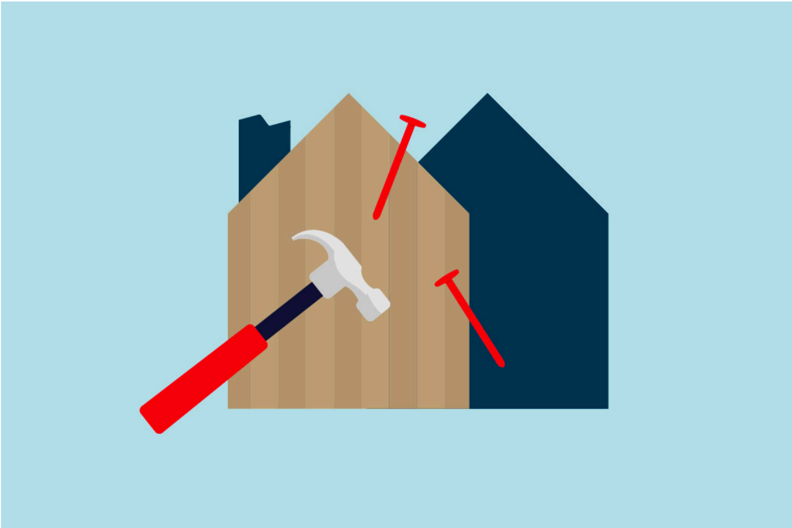
Ultimate Goal

Academic interest → New technology

Can we really build heart, package like DNA-packing, Engineered baby?

Yes. Understand the tiny parts and scale them up.

Note: This course is fairly fundamental/ introductory level.



Objective of this course :
Introduce you to the hammer, that is required to build a house.

At the end of the course you will realize :

* Known laws of physics and chemistry are good enough to understand biology
* Only difficulty is too many players.

Course syllabus (BT 101, Introductory Biology)

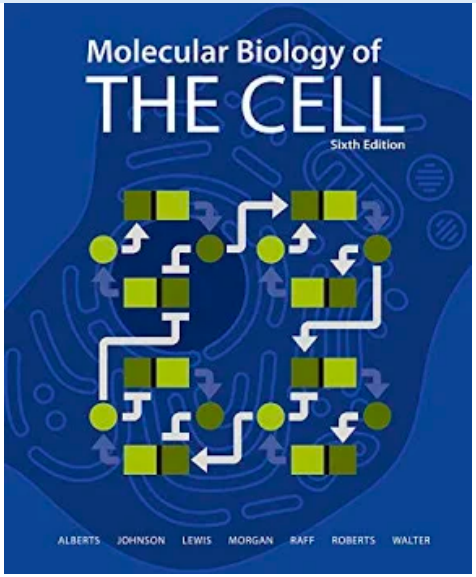
Nutrients, bioenergetics and cell metabolism: Essential nutrients to sustain life; biological energy and laws of thermodynamics, basics of aerobic and anaerobic glycolysis and citric acid cycle. (**Dr. Priyadarshi Satpati**)

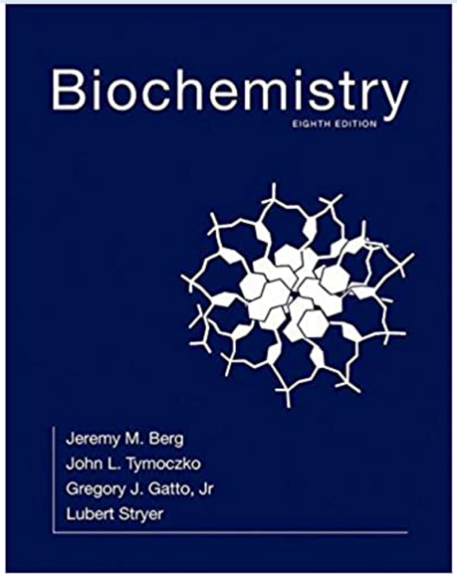
Genes and chromosomes: DNA, DNA replication; Central dogma of molecular biology: Transcription and translation; Mendelian Genetics; Genetic engineering/Cloning and its applications. (**Dr. Kusum K Singh**)

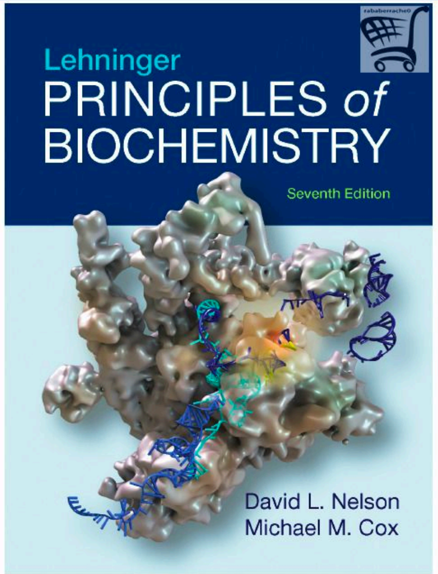
Evolution of life: Origin of Life; Darwin's concepts of evolution; Biodiversity. Cell, the structural and functional unit of life: Three domains of life; cell types, cell organelles and structure; Basic biomolecules of cell. (**Dr. Rajkumar P Thummer**)

Biological systems: Body systems required to sustain human physiology, special sense organs including hearing, taste, smell and visual receptors. (**Dr. Souptick Chanda**)

Text Books







References

1.N. Hopkins, J. W. Roberts, J. A. Steitz, J. Watson and A. M. Weiner, Molecular Biology of the Gene, 7th Ed, Benjamin Cummings, 1987.

2. C. R. Cantor and P. R. Schimmel, Biophysical Chemistry (Parts I, II and III), W.H. Freeman & Co., 1980.

3. C. C. Chatterjee, Human Physiology, Vol 1 & 2, 11th Ed, Medical Allied Agency, 1987.

4. Hall, B.K., Evolution: Principles and Processes, 1st Ed, Jones & Bartlett, 2011.

Time line

First Half
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Date: 05 Jan 2024 - 23 Feb 2024

Marks: 50

Mid-Sem : 26 Feb 2024, Monday

Instructors:

- Dr. P Satpati
- Dr. Kusum K Singh

Second Half
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Date : 04 March 2024 - 26 April 2024

Marks : 50

End-Sem : 01 May 2024, Wednesday

Instructors:

- Dr. Rajkumar P Thummer
- Dr. Souptick Chandra

Dr. Priyadarshi Satpati

Dr. Kususm K Singh

Dr. Rajkumar P Thummer

Dr. Souptick Chanda

Grading = 100 marks = MidSem (50) + EndSem (50)

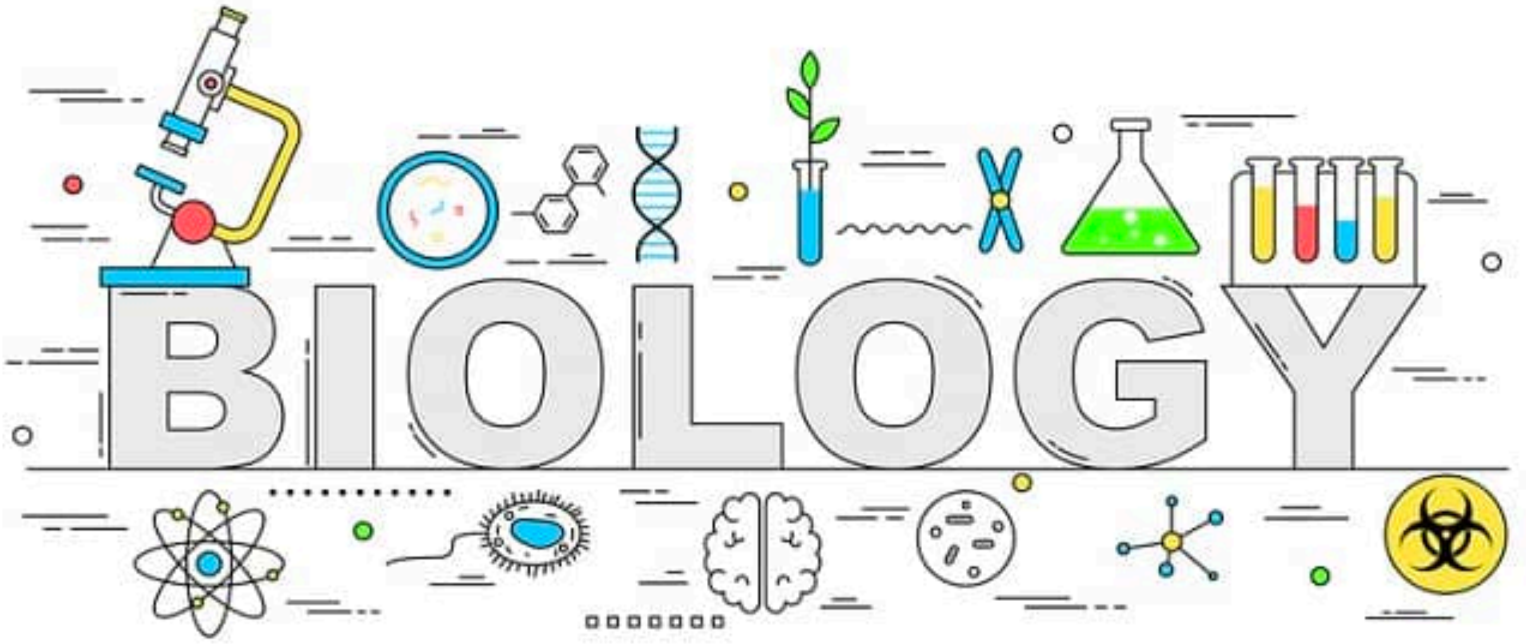
Format of the course

- PDF file will be uploaded after the instruction (class time-table).

- Every Lecture will be divided into two parts. Every part will be followed by discussion in the class.

TIPS

- **Keep paper and pen with you.**
- **Write down the doubts (if any) and the slide number**
- **We will discuss this in every class.**



Next: Key concepts of Bioenergetics